Context Compression for LLM Agents: A Survey of Methods, Failure Modes, and Evaluation

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Paper List

Abstract

As Large Language Models (LLMs) evolve into autonomous agents for long-horizon tasks, managing unbounded interaction trajectories under fixed context budgets becomes a core systems challenge. Unlike standard long-context documents, agent trajectories are heterogeneous and interleave observations, reasoning traces, and tool executions, so compression must preserve temporal dependencies, actionable state, and structural fidelity. Yet existing methods remain fragmented, making it difficult to compare design choices and reason about their reliability implications. This survey introduces a unified taxonomy of agent context compression along three dimensions: *compression target* (what is compressed), *compression mechanism* (how it is transformed and retained), and *control policy* (who decides when compression is triggered). We further organize recurring failures in compressed execution into **F1**: *Pre-compression Decision Error*, **F2**: *In-compression Information Loss*, and **F3**: *Post-compression Access Failure*, and examine domain-specific trade-offs in software engineering, web navigation, and deep research. By unifying the design space, failure taxonomy, and evaluation perspective, this survey provides a foundation for building scalable and recoverable LLM agents.

1 Introduction

With the rapid evolution of LLM agents, long context has emerged as a critical challenge (Du et al., 2025b; Bai et al., 2024; Wang, 2025; Wang et al., 2025a) across open-ended domains like automated software engineering (Xi et al., 2023; Shi et al., 2026c; Peng et al., 2026; Chen et al., 2025b), visual GUI navigation (Nguyen et al., 2025; Zhang et al., 2025a), and deep research (Zhang et al., 2025b; Huang et al., 2025). As agents continuously interact with dynamic environments, their operational history manifests as an **unbounded agentic trajectory**. This trajectory, typified by the **interleaved and heterogeneous** ReAct paradigm of Actions, Thoughts, and Observations (A-T-O) (Yao et al., 2023), rapidly exhausts the LLM's context window to trigger a severe ***context explosion***. Such an explosion precipitates a cascade of cognitive failures where information density plummets, critical constraints fade, and long-horizon planning collapses (Liu et al., 2024; Wang et al., 2024a, 2025c). Blindly pursuing longer context windows does not eliminate the need for compression, as models are inherently inept at effectively utilizing vast amounts of uncompressed, noisy textual information (Liu et al., 2024; Du et al., 2025b; Wang et al., 2025b), rendering **effective context management indispensable** for autonomy.

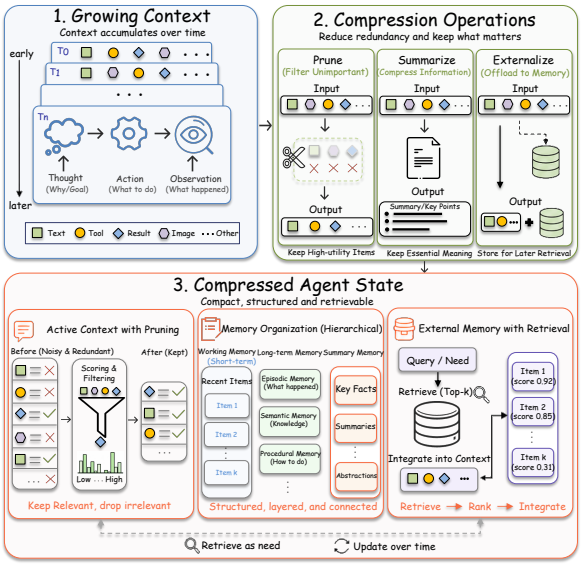


Figure 1: Illustration of the *context explosion* dilemma and the compression paradigm. Selective distillation transforms unbounded A-T-O sequences into structured representations while preserving operational fidelity.

Context compression emerges as a highly ef-

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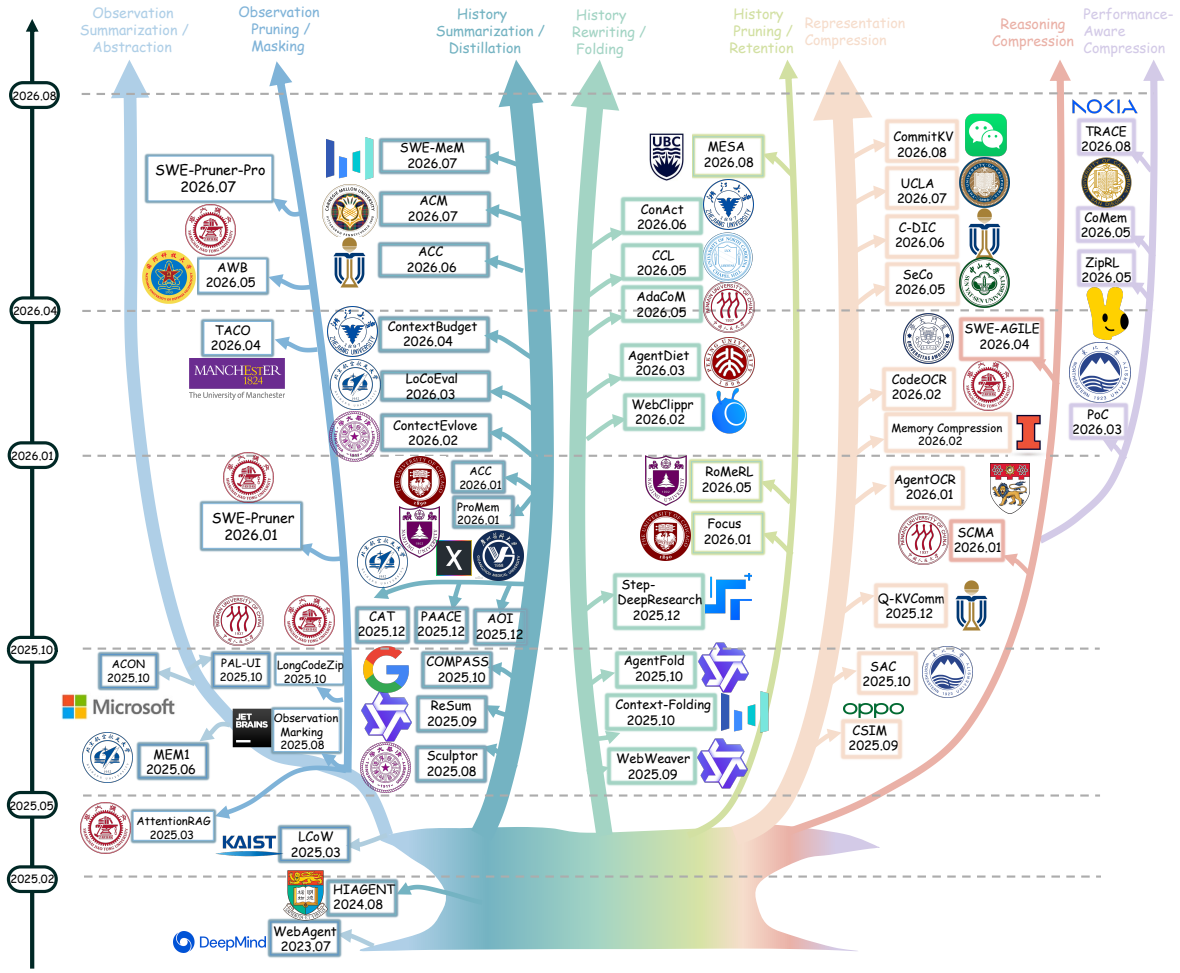


Figure 2: A design-space view of agent context compression. The figure synthesizes targets, mechanisms, and control policies into a method-level landscape, grouping approaches by compression semantics while preserving chronological evolution. See Appendix A and B for methodology, taxonomy details, and open-source resources.

fective pathway to mitigate this crisis, exemplified by real-world systems like ✳ Claude Code[1] that proactively compact contexts upon breaching predefined trajectory thresholds. Unlike the compression of static long documents explored in prior literature (Wang et al., 2026b; Li et al., 2024a; Xiao et al., 2024; Fang et al., 2025a), the profound **structural rigidity** of A-T-O sequences in agentic trajectories renders standard text reduction inapplicable. Consequently, as conceptualized in Figure 1, effective agent compression must **distill massive, noisy trajectories into compact, actionable states** while strictly **preserving causal action-observation dependencies** and structural integrity.

The prevalence of autonomous agents has catalyzed a rapid proliferation of agentic context compression techniques. These methods are scattered across different compression targets, transformation mechanisms, and control strategies, ranging from pruning and summarization to selective retrieval and representation-level interventions (Lindenbauer et al., 2025; Xiao et al., 2025; Wu et al., 2026c; Ye et al., 2025; Wu et al., 2026d,b; Feng et al., 2026; Wu et al., 2026a). Yet these efforts remain largely preliminary and disjointed explorations, lacking any **systematic synthesis** or **holistic perspective**. This conceptual fragmentation obscures a principled understanding of both the efficacy of underlying algorithmic designs and how specific compression choices trigger **distinct downstream agent failures** (Mei et al., 2025). Consequently, practitioners are left without a theoretical framework to navigate the complex trade-offs among **context footprint**, **structural fidelity**, and **historical recoverability**.

Prior surveys on prompt compression (Jiang et al., 2023; Li et al., 2025c) focus on the single-turn reduction of static inputs, whereas we study multi-turn trajectories whose relevance structure evolves over time. Work on KV cache (Li et al., 2024a; Liu et al., 2025e; Mathur et al.) and model

[1] ✳ Claude Code: https://claude.com/product/claude-code.

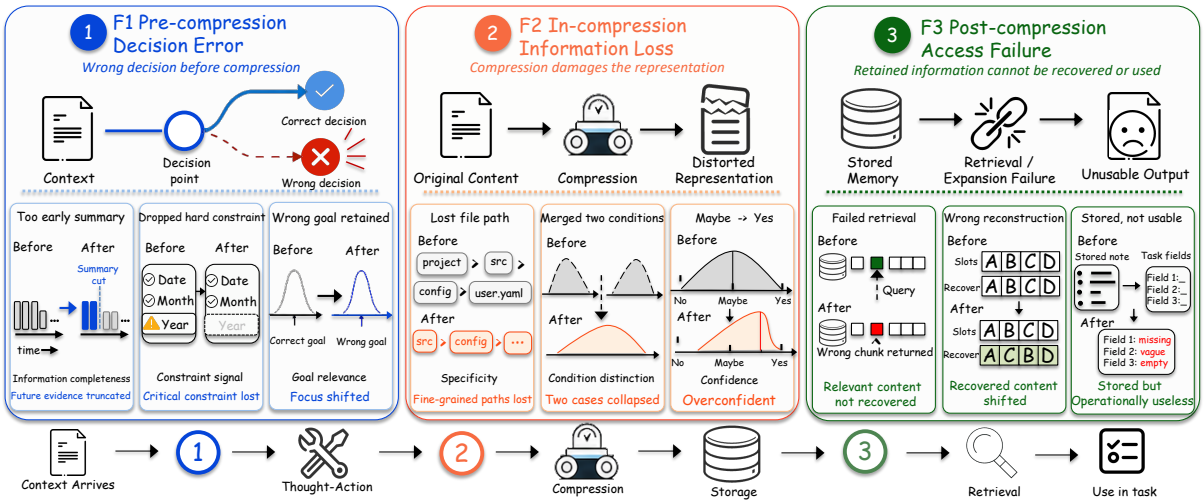


Figure 3: A temporal taxonomy of context compression failures. Failures are classified by the earliest point at which they occur in the compression pipeline.

compression (Zhu et al., 2024; Liu et al., 2025a; Xu and McAuley, 2023) addresses efficiency at the model or systems layer, while our concern is application-level context control during execution. Surveys on agent memory (Tang et al., 2026b; Zhang et al., 2025c; Hu et al., 2025c) organize methods by storage form, whereas we emphasize the intervention points, objects, and resulting forms of compressed context. Broader context engineering surveys span retrieval, memory, and orchestration at a high level (Mei et al., 2025); by contrast, we provide a focused treatment of compression as a **first-class operational capability** in agent systems.

As agents enter the era of long-horizon tasks, context compression has become a three-dimensional systems problem rather than a single-turn prompt optimization trick. Yet a systematic review of this emerging domain remains absent. To fill this gap, we introduce a unified taxonomy organized around three questions: ***compression target*** (what is compressed), ***compression mechanism*** (how it is transformed and retained), and ***control policy*** (who decides when compression is triggered). These dimensions map directly onto the operational pipeline: targets specify **Select**, mechanisms instantiate **Compress** and **Store**, and policies govern when these stages, and when needed **Recover**, are activated. We further formalize three recurring failure modes, **F1**: *Pre-compression Decision Error*, **F2**: *In-compression Information Loss*, and **F3**: *Post-compression Access Failure*, and use them to expose reliability trade-offs. Finally, we critique current evaluation paradigms (Liu et al., 2025c) and argue for metrics beyond end-task success, including information density, temporal consistency, and error propagation.

The rest of the survey is organized as follows. Section 2 introduces the background and unified pipeline, Section 3 presents a formal lens, Sections 4 and 5 develop the taxonomy and failure analysis, and the remaining sections discuss domain-specific trade-offs, evaluation, future directions, and conclusion.

2 Redefining Context: From Static Prompts to Dynamic Trajectories

2.1 What Is Unique About Agent Context

Formally, agent context is a time-indexed state C_t continuously updated by interaction tuples $x_t = (a_t, t_t, o_t)$ of actions, thoughts, and observations (Yao et al., 2023). Unlike standard long-text processing, this dynamic trajectory exhibits four fundamental properties that distinguish it from static prompts:

Property 1: Dynamic Growth. Context is not provided all at once but expands incrementally by absorbing new interactions x_t with each agent step. Unlike static documents with fully exposed structures, agentic compression operates blindly on this continuous stream, dynamically distilling history without foreseeing future states.

Property 2: Heterogeneous Composition. Context comprises highly distinct structured content, including natural language plans, code snippets, JSON tool outputs, screenshots, and error logs. While natural language thoughts (t_t) may endure lossy abstraction, actions and observations often demand strict syntactic fidelity, mandating type-

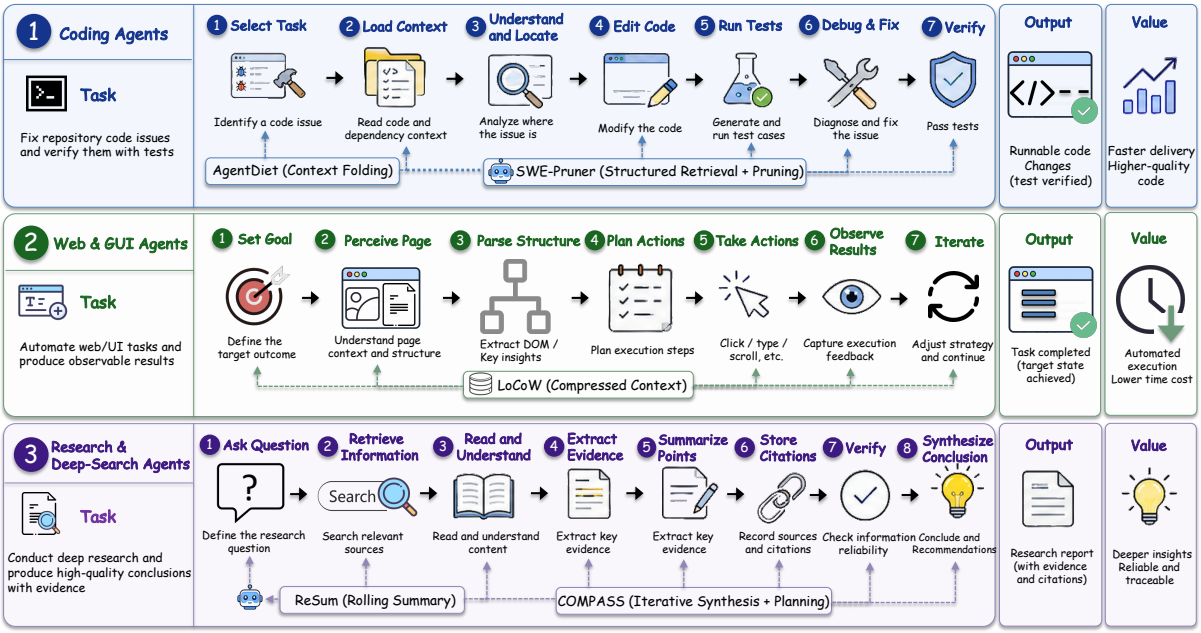


Figure 4: Domain-specific context compression regimes, contrasting setup, workflow, interaction loop, dominant risks, and strategy choices.

aware rather than uniform compression.

Property 3: Multi-step Dependency. Information compressed or discarded at a certain step may only be needed several steps later. Driven by continuous streaming, the utility of any specific tuple component e manifests over future steps: $\mathcal{U}(e) = \mathbb{E}\left[\sum_{\tau=t}^{T} \gamma^{\tau-t} \cdot u_\tau(e)\right]$. Consequently, it is inherently difficult to predict future demands during compression, as the true importance of e remains unobservable at time t.

Property 4: Error Propagation. Errors or omissions introduced by compression are not discovered locally. Instead, a state distortion $\delta_t = d(C_t, \tilde{C}_t)$ propagates backward along the trajectory: $C_{t+k} = U^{(k)}(\tilde{C}_t, x_{t+1:t+k})$. These errors accumulate irreversibly and are treated as factual grounding by subsequent reasoning steps, permanently shifting the action space and derailing the agent from the optimal path (Wang et al., 2024b).

► ***Why Traditional Compression Fails*** ◄ Traditional methods satisfy one-off budget constraints by optimizing for immediate relevance (Jiang et al., 2023, 2024). For example, KV cache compression typically evicts tokens with low current attention scores (Xiao et al., 2024; Li et al., 2024d). However, under the four properties above, applying such static interventions to an evolving agentic loop triggers compounding distortions, as continuous interaction defies any static definition of information importance.

2.2 A Unified Operational Pipeline for Agent Context Compression

We formulate a unified four-stage pipeline that decomposes compression into an operator system over the context lifecycle:

$$C_{t+1} = \underbrace{R(q_{t+1}, \mathcal{M}_t)}_{\text{Recover}} \oplus \underbrace{M\Big(\Phi\big(S(C_t)\big)\Big)}_{\text{Store}\leftarrow\text{Compress}\leftarrow\text{Select}} \oplus x_{t+1} \tag{1}$$

- **Select (S):** A critical triage phase where the system determines which subsets of the context require compression and which demand exact preservation.
- **Compress (Φ):** Specific reductive or structural transformations, ranging from simple truncation to latent encoding, are executed on the selected subset.
- **Store (M):** The resulting compressed representations are committed either to the active working memory or offloaded to an external storage state $\mathcal{M}_t$.
- **Recover (R):** Driven by the necessity of deferred utility, relevant compressed information is selectively retrieved and decoded from $\mathcal{M}_t$ using the current query q_{t+1}.

3 Compression as Sufficient State Approximation

We argue that agent context compression is best understood as a sufficient-state approximation prob-

lem: the goal is to maintain the shortest representation that remains sufficient for future action.

Our formalization is inspired by ACON (Kang et al., 2025); see Appendix E for the distinctions.

Compression as State Transformation. Let C_t denote the current context state. We treat each compression method as a transformation

$$O_i : \mathcal{C} \to \mathcal{C}, \qquad C'_t = O_i(C_t),$$

where O_i maps C_t to a shorter, reorganized, or externally mediated representation C'_t under task-specific constraints. This abstraction is deliberately broad. It covers the operator families discussed later in the survey, including truncation, pruning, summarization, externalization, and representation-level compression.

Ideal Sufficient Memory. From an algorithmic-information perspective, these operators can be viewed as practical approximations to an ideal sufficient memory. Let E_t denote the currently accessible environment state, A the agent policy or executor, τ the task or task distribution, and ϵ the maximum acceptable degradation in downstream performance. We define the ideal sufficient memory length as

$$L_\epsilon(C_t) = \min_{\tilde{C}_t \in \mathcal{S}_t} K(\tilde{C}_t \mid E_t) \quad \text{s.t.} \quad \Delta_\tau(A; C_t, \tilde{C}_t, E_t) \leq \epsilon, \tag{2}$$

where $\mathcal{S}_t = \mathcal{S}(C_t, E_t)$ denotes the admissible space of compressed states derived from C_t under environment E_t. Here $K(\tilde{C}_t \mid E_t)$ is standard conditional Kolmogorov complexity, and $\Delta_\tau(A; C_t, \tilde{C}_t, E_t)$ measures the loss in future task performance induced by replacing C_t with $\tilde{C}_t$. The quantity L_ϵ is not meant as a computable training objective; rather, it serves as a stylized lower bound. A compressed state is useful only insofar as it remains compact, task-sufficient, and recoverable for future decisions.

From Approximation to Failure. This objective turns the next two sections into one question: how does a method approximate a sufficient future state, and where can that approximation fail? Truncation bets that dropped spans have no future value. Pruning preserves an exact sufficient subset. Summarization replaces evidence with an executable abstraction. Externalization keeps part of the sufficient state off-context and shifts the bottleneck to recovery. Representation-level compression pushes sufficiency into latent space and sacrifices transparency. Under this view, the taxonomy compares approximation strategies, while the failure modes identify where sufficiency or recoverability is lost.

4 A Taxonomy of Agent Compression

4.1 From Pipeline to Design Space

The unified pipeline in § 2.2 can be read as a comparative design space for agent compression. We organize the literature along three axes: *what* is selected for compression, *how* it is transformed, and *who* decides when compression occurs. These axes map naturally onto the pipeline: targets define the input to **Select** (S), mechanisms implement **Compress** (Φ) and **Store** (M), and control policies govern when these stages, and when necessary **Recover** (R), are invoked.

4.2 Compression Targets (*What*)

We begin with the objects processed by the **Select** (S) operator. Table 1 summarizes these targets in terms of granularity, information density, causal coupling, and recoverability. Together, these properties determine both compressibility and likely failure symptoms.

This section adopts the definitions of failure modes detailed in Section 5, including **F1**: *Pre-compression Decision Error*, **F2**: *In-compression Information Loss* and **F3**: *Post-compression Access Failure*.

4.2.1 Observation Compression

Observation compression targets raw inputs, such as tool outputs, HTML DOMs, logs, or screenshots. These inputs are high-volume and redundant, but a single omitted selector, trace line, or visual cue may determine the next action. Methods such as The Complexity Trap (Lindenbauer et al., 2025) therefore treat it as a low-cost front-end intervention, typically via masking or shallow filtering.

4.2.2 Trajectory Compression

Trajectory compression focuses on accumulated multi-step histories, including failed attempts, redundant tool calls, and repetitive loops. Its main difficulty is deferred utility: information that appears irrelevant at step t may become indispensable later. As a result, trajectory compression is especially vulnerable to **F2**: *In-compression Information Loss* and delayed downstream inconsistency. Representative approaches include controller-mediated re-

Target	Granularity	Information Density	Causal Coupling	Recoverability	Typical Failure Symptom
Observation	Token / Segment	★★	★★	★★★	Structural Distortion
Trajectory	Episode	★	★	★★	Planning Drift
Plan & Reasoning	State	★★★	★★★★	★	Metacognitive Error
Memory State	State	★★★	★★★	★	Retrieval Blindness
Representation	Token / Latent	★★★★	—	—	Semantic Collapse

Table 1: A comparative anatomy of compression targets. Each target is characterized by its granularity, information density, causal coupling, and recoverability, which collectively shape its compressibility and its failure symptom.

duction in AgentDiet (Xiao et al., 2025), periodic summarization in ReSum (Wu et al., 2026c), and proactive folding in AgentFold (Ye et al., 2025).

4.2.3 Plan and Reasoning Compression

Plan and reasoning compression targets internal artifacts, including subgoals, deliberative traces, self-reflections, and task decompositions (Zeng et al., 2025; Zhang et al., 2026e). These representations are compact but brittle: dropping a seemingly minor premise can invalidate an entire downstream plan. This target is therefore sensitive to **F1**: *Pre-compression Decision Error*, because the system must correctly judge which reasoning artifacts remain decision-critical. Representative methods include feedback-driven summarization in ACON (Kang et al., 2025) and agent-initiated memory management in Focus (Verma, 2026).

4.2.4 Memory State Compression

Memory state compression maps an unbounded interaction stream into a fixed-capacity representation that survives beyond the immediate context window. It most explicitly realizes the **Store** (M) stage, so its main bottleneck is accessibility rather than faithfulness. Once information is offloaded, the system must still recognize when that memory matters and reconstruct it at use time. Consequently, memory-state compression is most directly exposed to **F3**: *Post-compression Access Failure*. Learned memory policies such as MEM1 (Zhou et al., 2025b) address this challenge by coupling state formation with future retrieval utility.

4.2.5 Representation-Level Compression

Representation-level compression operates below the natural-language interface, targeting KV caches, multimodal token embeddings, or other latent states (Shi et al., 2026b; Ma et al., 2026). It often delivers the strongest efficiency gains, but with minimal recoverability and weak inspectability. These methods therefore shift the problem from semantic editing to opaque state management. Representative examples include AgentOCR (Feng et al., 2026), Q-KVComm (Kriuk and Ng, 2025), and SAC (Liu et al., 2026b).

4.3 Compression Mechanisms (*How*)

Having identified the targets, we now turn to the transformations themselves. What most distinguishes compression mechanisms is how they relate the compressed result to the original context. Some remove content, some paraphrase it, some retain an exact subset, and others store information so that it can be recovered later. This distinction is crucial because methods with similar compression ratios may differ sharply in *faithfulness*, *inspectability*, and *downstream recoverability*.

4.3.1 Masking and Truncation

These operators keep a prefix or suffix of the context, or selectively mask designated spans:

$$O_{\text{trunc}}(\mathcal{C}) = \mathcal{C}[:k],\ O_{\text{mask}}(\mathcal{C}) = \text{mask}(\mathcal{C}, S).$$

These operators, which directly modify the support of C through span removal or placeholder substitution, are universal, low-overhead, and easy to deploy. That same directness also defines their limitation: once content is removed, recoverability is lost unless another subsystem stores it elsewhere. Such operators are therefore best suited to settings in which marginal utility decays rapidly with recency or position, as in sliding-window baselines or hard budget enforcement (Wang et al., 2025d).

4.3.2 Summarization and Abstraction

These operators rewrite a long context into a shorter semantic representation:

$$O_{\text{sum}}(\mathcal{C}) = \hat{C},\ |\hat{C}| \ll |\mathcal{C}|.$$

Unlike truncation, summarization preserves utility by semantic rewriting rather than direct deletion. This often yields substantially higher information density and better long-horizon compactness, which explains its popularity in research and search agents. The trade-off is that faithfulness becomes model-dependent: once context is rewritten,

omissions, over-generalizations, and spurious inferences may be introduced, making summarization a primary locus of **F2**: *In-compression Information Loss*. Representative systems include LCoW (Lee et al., 2025), COMPASS (Wan et al., 2025), and ReSum (Wu et al., 2026c).

4.3.3 Pruning and Reduction

Pruning removes content selectively while preserving the remaining structure exactly:

$$O_{\text{prune}}(\mathcal{C}) = \mathcal{C} \setminus R,$$

where R is a subset selected by a relevance, dependency, or structural criterion (Fang et al., 2025a; Shi et al., 2025). Unlike summarization, pruning leaves retained evidence untouched. It is therefore especially valuable when syntactic form, pointer alignment, or execution traces must remain exact, as in coding and GUI settings. In exchange, pruning typically compresses less aggressively than summarization unless the context contains substantial redundancy. Representative examples include SWE-Pruner (Wang et al., 2026c,d) and WebClipper (Wang et al., 2026a).

4.3.4 Externalization and Retrieval

This mechanism splits context into a working state and an external store:

$$O_{\text{ext}}(\mathcal{C}) = (C', M),\ O_{\text{ret}}(\mathcal{C}, M) = \mathcal{C} \oplus r(M).$$

It is the clearest realization of **Store** (M) and **Recover** (R) in the pipeline. By turning part of compression into retrieval, it permits aggressive footprint reduction so long as critical information can later be reconstructed on demand. It is therefore the only mechanism that combines high compression with principled recoverability. Its main risk shifts toward **F3**: *Post-compression Access Failure*: not corruption, but missing, mistimed, or incorrect reaccess. Systems such as Step-DeepResearch (Hu et al., 2025a), CAT (Liu et al., 2025b), and WebWeaver (Li et al., 2025b) rely heavily on this paradigm.

4.3.5 Representation Compression

Operating on latent structures, $O_{\text{repr}}(\mathcal{C}) = \tilde{C}$ compresses history without explicit textual rewriting. These methods optimize efficiency most directly, but they do so by moving compression beneath the symbolic interface available to the agent and the researcher. As a result, they offer limited interpretability, weak controllability, and little explicit support for exact recovery. Methods such as DebateOCR (Wu et al., 2026a) illustrate this trade-off.

4.4 Control Policies and Intervention Timing (*Who* and *When*)

Control policy, not compression mechanism, determines when intervention occurs. The same operator may be triggered reactively at a budget threshold, periodically by an external controller, or proactively by the agent itself. We therefore separate operator family from control policy: policy specifies when **Select**, **Compress**, and, where applicable, **Recover** are invoked, who makes that decision, and which signals are sufficient to trigger action.

4.4.1 System-Controlled Policies

System-controlled policies apply fixed rules to trigger compression:

$$\pi_{\text{sys}} : \mathcal{C} \mapsto (i, \theta).$$

Depending only on observable signals such as length, position, or step count, they are predictable, cheap, and easy to benchmark. Their weakness is semantic blindness: they intervene because a budget is reached, not because task utility has changed. Such policies therefore dominate early and production-constrained systems, including predefined schedules in AgentDiet (Xiao et al., 2025) and threshold-based compaction regimes.

4.4.2 External Controller Policies

External-controller policies delegate the decision to a separate module:

$$\pi_{\text{ext}} : \mathcal{C} \to (i, \theta).$$

The controller may be a smaller model, a planner, or a retrieval manager that observes the evolving context and selects a compression action for the main agent. This decoupling improves modularity and sometimes yields more stable control, because compression can be optimized independently of task execution. The cost is additional latency, token overhead, and a new interface on which information can be lost. Examples include PAL-UI (Liu et al., 2025f) and AOI (Bai et al., 2026).

4.4.3 Agent-Controlled Policies

Agent-controlled policies make compression part of the agent's own action space:

$$\pi_{\text{agent}}(C_t, s_t) \to (i, \theta),$$

where s_t denotes the agent's internal state. This policy class is the most semantically aware, because the same system that reasons about the task also decides which context should be compressed, preserved, or revisited. It is therefore well suited to proactive memory management and self-reflective

System	Select (S)	Compress (Φ)	Store (M)	Recover (R)
Claude Code	Reactive, threshold-based selection.	Auto-compaction of prior interaction history.	Compacted state retained in active context.	Implicit recovery via continued execution.
Cursor	Retrieval-first selection of files, code blocks, and rules.	Relevance filtering and contextual assembly.	Repository index, project rules, and workspace context.	Re-injection of retrieved files, rules, and evidence.
Codex	Task-grounded selection over files, instructions, and threads.	Session continuation and workspace grounding.	Task threads, instruction files, and repository state.	Recovery through continued threads and file re-access.
Pi	Threshold; safe cut; tail budget; tool-pair preservation	LLM summary; iterative update	JSONL tree; metadata; memory index	Summary substitution; rebuild; replay
Hermes	Multi-gate trigger; head/tail protection; tool pruning	LLM summary; pruning; fallback	SQLite/FTS5; archive; memory providers	Prompt rebuild; retrieval; context injection

Table 2: Pipeline-level comparison of recent production-grade and open-source coding agents, organized by the stages in Section 2.2.

intervention, as in CAT (Liu et al., 2025b), Focus (Verma, 2026), and ContextBudget (Wu et al., 2026e). Its weakness is precisely the agent's own fallibility: errors in self-assessment readily manifest as **F1**: *Pre-compression Decision Error*.

4.4.4 Learned Compression Policies

Learned policies optimize compression behavior from data:

$$\pi_{\text{learn}} = \arg\max_{\pi} \mathbb{E}_{\tau\sim\pi}\left[R_{\text{task}}(\tau) - \lambda\,\text{Cost}(\tau)\right].$$

Whether trained by supervised fine-tuning or reinforcement learning, they seek to align compression decisions with downstream task return rather than handcrafted heuristics. This makes them the most expressive policy class and, in principle, the closest approximation to the ideal sufficient-memory objective in Section 3. In practice, however, they are also the most resource-intensive and the most sensitive to reward misspecification. Representative examples include ACON (Kang et al., 2025), MEM1 (Zhou et al., 2025b), and related policy-learning approaches.

5 Failure Modes

We organize context compression failures by the earliest stage at which they arise: **F1**: *Pre-compression Decision Error*, **F2**: *In-compression Information Loss*, and **F3**: *Post-compression Access Failure*, as shown in Figure 3.

F1: *Pre-compression Decision Error* refers to failures that arise before compression itself, when the system chooses the wrong moment, target, or granularity of compression. The issue is not the compression operator, but a pre-compression decision that already departs from task requirements. Typical cases include premature compression (Yang et al., 2026a; Liu et al., 2025b), misjudging the retention value of critical information (Liang et al., 2026; Feng et al., 2026), or compressing constraints that should remain exact (Liu et al., 2025b). A representative case, detailed in Appendix D, removes authentication-related state before the agent reaches the step where it becomes necessary.

F2: *In-compression Information Loss* refers to failures introduced by the compression operation itself, where representation transformation corrupts semantics, structure, relations, constraints, or fine-grained detail. The central question is whether compression preserves the original information faithfully enough for later use. Typical cases include omitted semantics (Chen et al., 2026a; Yang et al., 2026a), disrupted relational structure (Wu et al., 2026d), oversimplified constraints (Chen et al., 2026a; Liu et al., 2025b), or compressed uncertainty that becomes an unwarranted conclusion (Yang et al., 2026a). Appendix D includes a trajectory-level case in which compression preserves the broad task intent but loses a fine-grained constraint, leaving the later agent state semantically incomplete.

F3: *Post-compression Access Failure* refers to failures of post-compression accessibility: information is retained in some form, but cannot be correctly retrieved, expanded, or reconstructed when needed. The issue is therefore recoverability rather than pre-compression choice or in-compression corruption. Typical cases include failed retrieval (Li et al., 2026a; Zeng et al., 2026; Chen et al., 2026a), incorrect reconstruction (Chen et al., 2026a), or compressed states that remain storable yet functionally unusable (Li et al., 2026a; Zeng et al., 2026). The Appendix D case makes the same point from the

access side: the correct compressed memory exists, but later retrieval returns a semantically similar yet wrong memory item.

Together, **F1–F3** define a temporal failure taxonomy over the compression pipeline. For hybrid cases, we assign the label to the earliest failure that causally triggers later errors, since agent workflows naturally exhibit error propagation (Wang et al., 2024a).

6 Domain-Specific Analysis

Compression is domain-dependent: coding agents require structural fidelity and state continuity; web and GUI agents must preserve heterogeneous multimodal observations; DeepResearch agents depend on deferred evidence recovery across long horizons. Figure 4 summarizes these regimes.

Coding Agents Coding places the strongest pressure on structure preservation, exact evidence retention, and reliable recovery (Hu et al., 2026a; Zhang et al., 2026c). Accordingly, methods such as SWE-Pruner (Wang et al., 2026c), LongCodeZip (Shi et al., 2025), and CAT (Liu et al., 2025b) emphasize structure-aware pruning and explicit state retention. Table 2 compares recent production-grade coding agents through the unified operational pipeline, using only public evidence and high-confidence product signals. The contrast is instructive. Claude Code emphasizes **Compress** through reactive compaction, Cursor places more weight on **Store** and **Recover** through repository indexing and retrieval, and Codex appears most grounded in **Select** plus workspace continuation. Across all three, the common pattern is not aggressive lossy shortening, but coordinated pipeline control under tight context budgets.

Web & GUI Agents Web and GUI agents operate over heterogeneous, partially redundant observations. Methods such as LCoW (Lee et al., 2025) and PAL-UI (Liu et al., 2025f) exploit redundancy, but representation-level compression must still preserve modality and interface cues.

DeepResearch Agents Research and deep-search agents are defined less by structural fidelity than by delayed evidence use. Their main risk is that abstraction hides evidence that matters only later. ReSum (Wu et al., 2026c) and COMPASS (Wan et al., 2025) favor summarization, whereas Step-DeepResearch (Hu et al., 2025a) and WebWeaver (Li et al., 2025b) externalize or retrieve evidence to preserve long-horizon dependencies.

7 Beyond Task Success: Rethinking Evaluation

Outcome-Centric Evaluation. Benchmarks still judge context compression mainly by downstream task success. Success rate, with token count or compression ratio as an auxiliary proxy, remains the default protocol. This is convenient but inadequate: success entangles compression with planning and execution, while shorter prompts reveal little about whether task-critical evidence survives. Cross-paper comparison is further blurred by changes in backbone, task set, and context budget.

The Missing Object. The missing object is not the task outcome but the compressed state itself. For agents, four dimensions are decisive: *information density*, *recoverability*, *error propagation*, and *controller overhead*. Together, they ask whether compression preserves task-critical evidence, supports later restoration, avoids long-horizon failure amplification, and remains efficient enough to justify its control cost.

From Outcomes to State Quality. We therefore evaluate compressed states through $Q = (D, R, P, O)$, where D denotes task-relevant density, R recoverability, P propagation risk, and O operational overhead, and summarize comparison through $J = \alpha D + \beta R - \gamma P - \delta O$. The point is simple but consequential: compression should be judged as state quality, not only task outcome. This reframing ties benchmark outcomes to compression quality under domain-specific constraints $\mathrm{Constraint}_i$. Taken together, this reframing motivates a broader discussion of recoverable, faithful, and long-horizon context management, which we defer to Appendix E.

8 Conclusion

Agent context compression is no longer just a preprocessing convenience. For long-horizon agents, it is a state-management problem over a changing workspace, not a one-shot prompt-shortening step. This survey structures the area along three axes: targets, mechanisms, and control policies. It also introduces a failure taxonomy and an evaluation framework that goes beyond task success. The key challenge is not only stronger compression, but compression that remains recoverable, faithful, and aligned with long-term agent execution.

Limitations

Coverage. Because agent context compression is evolving rapidly, our synthesis is intended to be representative rather than exhaustive, especially for the newest systems and engineering practices still stabilizing in the literature.

Scope. We deliberately focus on context compression during agent execution, rather than adjacent areas such as general long-context modeling, context engineering, memory management, or model-level KV-cache optimization, except where they directly inform agent compression. We view these neighboring directions as natural extensions and leave a unified treatment to future work.

Abstraction Level. Our taxonomy prioritizes conceptual unification across targets, mechanisms, policies, failure modes, and evaluation. This improves comparability, but necessarily abstracts away some implementation-level variation across backbone models, task settings, and reporting protocols.

We view these limitations primarily as boundaries of scope rather than weaknesses of the framework, and hope the structure offered here provides a stable basis for more exhaustive updates as the field matures.

AI Usage Disclosure

The authors used AI-based writing assistants to improve grammar, phrasing, and local clarity during manuscript preparation. All AI-assisted suggestions were reviewed, revised, and selectively incorporated by the authors, who take full responsibility for the final text, technical framing, citations, and conclusions.

Ethical Statement

This work is a survey of publicly available research papers, technical reports, product documentation, and open-source resources on context management for LLM agents. It does not involve private data, human subjects, or sensitive personal information. We aim to represent prior work accurately and cite sources faithfully; any discussion of deployed systems is based only on publicly accessible evidence and is used solely for scholarly analysis.

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A Methodology of the Survey

A.1 Search Strategy

Because agent context compression is an emerging topic without a unified terminology, we adopted a broad search strategy rather than relying on a single keyword. We searched ACL Anthology, DBLP, arXiv, Google Scholar, Semantic Scholar, and major NLP, machine learning, software engineering, and agent-system venues. The search covered both explicit compression terms and adjacent terms used in related communities.

Specifically, we used keyword combinations such as *"LLM agent"* with *"context compression"*, *"trajectory compression"*, *"history summarization"*, *"context pruning"*, *"context folding"*, *"agent memory compression"*, *"memory retrieval"*, and *"long-horizon agent context"*. We also searched domain-specific terms, including *"coding agent context"*, *"web agent memory"*, *"GUI agent observation compression"*, and *"deep research agent memory"*. These queries were designed to capture papers that may not use the exact phrase *context compression* but nevertheless reduce, rewrite, externalize, or retrieve agent history under a limited context budget.

In addition to keyword search, we performed backward and forward citation tracing from representative work on prompt compression, long-context modeling, agent memory, web agents, coding agents, and deep-research agents. This step was necessary because many relevant systems describe compression-related operations as *memory management*, *state retention*, *context engineering*, *trajectory summarization*, or *retrieval-based continuation*. We therefore treated terminology flexibly and focused on whether a method changes what historical information remains available to the agent during future execution.

A.2 Inclusion and Exclusion Criteria

We included a paper or system if it satisfied at least one of the following conditions. First, it proposes a method that compresses, filters, summarizes, prunes, folds, externalizes, or retrieves the context of an LLM agent during multi-step execution. Second, it introduces a memory or retrieval mechanism whose purpose is to preserve task-relevant information under a finite context budget. Third, it analyzes long-horizon agent behavior in a way that reveals failures caused by context reduction, memory selection, retrieval, or compressed-state reuse. Fourth, it describes a deployed or production-grade agent system with publicly documented context-management behavior relevant to the operational pipeline studied in this survey.

We excluded work whose primary contribution lies outside application-level agent context management. In particular, we excluded methods that only compress static, single-turn prompts unless they are explicitly adapted to multi-step agent trajectories. We also excluded model-level compression methods, such as weight quantization, model pruning, distillation, and generic KV-cache optimization, when they do not alter the information available to the agent as part of its executable context. Similarly, general retrieval-augmented generation was not treated as agent context compression unless retrieval is coupled to an evolving agent state, trajectory history, or memory store.

For borderline cases, we used a functional criterion: a work was considered relevant if its method changes how past observations, actions, thoughts, plans, or memory states are preserved and reused for future decisions under a context constraint. This criterion allowed us to include systems that do not explicitly use the term *compression* but still perform compression-like state management. When a method combined multiple components, such as summarization plus retrieval or pruning plus external memory, we categorized it according to the component that most directly determines the compressed state available to later agent execution.

Finally, for production-grade systems and technical reports, we only relied on public evidence and high-confidence descriptions. When implementation details were unavailable, we characterized the system at the level of observable pipeline behavior rather than making assumptions about proprietary mechanisms.

B Extended Taxonomies and Resources

B.1 Details about taxonomy

In the main text, we characterize context management methodologies based on three core dimensions, specifically *when*, *what*, and *how*, while further analyzing their application *domain* as a complementary perspective. Table 7 below instantiates this framework at the level of individual methods. Specifically, the *when* column indicates the trigger mechanism and decision maker for context compression or management, including system-controlled, external-controller, agent-controlled, and learned policies; the *what* column identifies

the context object being managed, such as observations, trajectories, memory states, plans and reasoning traces, or representation-level signals; and the *how* column summarizes the compression mechanism, including truncation and masking, pruning and reduction, summarization and abstraction, externalization and retrieval, and representation-level compression. In addition to these three dimensions, we report the *domain* of each method to indicate its primary task setting, and we analyze its distribution and characteristics separately in the main text. Taken together, the table provides a fine-grained view of how different methods are positioned within the taxonomy and how their design choices vary across task domains.

B.2 Open-Source Standouts

To more comprehensively reflect the openness and reproducibility of research in this area, we further collected and summarized the open-source methods included in this survey, as shown in Table 8. The table exclusively incorporates methods accompanied by publicly available code, presenting them across three columns: method name, open-source link, and publication year. This organization enables readers to efficiently locate the corresponding implementations and repository resources. As shown in the table, the current open-source work is concentrated in a small number of representative methods, and the release years span 2025 to 2026, which indicates that this field is still evolving rapidly and that its open-source ecosystem is steadily maturing. Overall, this statistics not only improves the reproducibility of our survey results, but also provides a direct entry point for future researchers to conduct method comparison, code replication, and further development.

B.3 Datasets & Benchmarks

Table 9 summarizes the representative datasets and benchmarks collected in `datasets_info.md`. We organize them by task family and retain only the information most useful for a survey on agentic context management: benchmark purpose, publication venue/year, and public availability. To keep the table compact, we move the task-focus and scale description into the benchmark cell itself, so each row reads as a short self-contained summary.

The *Benchmark* cell includes the dataset name, a citation to the original paper, and a brief description of the task and scale. The *Source and year* column records the original publication venue or release year, while the *Open Source* column provides the public repository or dataset URL when available, and indicates the absence of public releases otherwise. Benchmarks with closely related variants are grouped together to keep the presentation compact.

First, the selected benchmarks are intentionally diverse, encompassing software engineering, web browsing, office automation, mobile apps, deep research, long-context reasoning, memory, code generation, and long-document summarization. This diversity is essential, as context management methods frequently navigate the trade-off between retaining essential task-critical evidence and pruning redundant historical data, with individual benchmarks typically evaluating only a specific subset of these behaviors.

Second, numerous benchmarks are structured as closely related families or variants. For instance, *SWE-bench Verified* (Jimenez et al., 2024) and *SWE-bench Lite* (Jimenez et al., 2024) assess software issue resolution under different difficulty and edit constraints; *BrowseComp* (Wei et al., 2025), *BrowseComp-Plus* (Chen et al., 2025d), and *BrowseComp-ZH* (Zhou et al., 2025a) cover browsing-centric deep search in different settings; *Mind2Web* (Deng et al., 2023) and *Multimodal-Mind2Web* (Zheng et al., 2024) differ by adding visual grounding; *LongBench* (Bai et al., 2024), *LongBench V2* (Bai et al., 2025), *BABILong* (Kuratov et al., 2024), *NeedleBench* (Li et al., 2024c), and *NIAH* (Liu et al., 2024) probe long-context handling from complementary angles.

Third, in instances where a benchmark entry represents a comprehensive framework or family rather than an isolated dataset, we designate it by its representative name in the table and elucidate the included variants within the accompanying notes. This is especially helpful for benchmarks such as *MRQA* (Fisch et al., 2019), *Loghub* (Zhu et al., 2023), and *CompileBench / CRUST-Bench* (Khatry et al., 2025), where a single umbrella name aggregates multiple sub-datasets or tasks.

Finally, the *Open Source* column is interpreted broadly, providing the public dataset or repository URL when available, and explicitly noting cases where no public release could be identified within the collected sources. The original publication year and venue are preserved in the *Source and year* column.

C Implementation Details of Compression Operators

This appendix supplements the taxonomy in Section 4.3 by summarizing concrete implementation details of representative compression operators. We focus on three methods with relatively explicit descriptions: ReSum (Wu et al., 2026c) and AgentFold (Ye et al., 2025) for summarization and abstraction, and SWE-Pruner (Wang et al., 2026c) for pruning and reduction. The goal is not to reproduce full prompts verbatim, but to identify the prompt roles, response templates, algorithmic steps, and compression-specific design choices that instantiate the operator families discussed in the main text.

C.1 Prompts for Summarization and Abstraction

Summarization and abstraction methods rewrite long interaction histories into shorter executable states. Table 3 compares the concrete prompt or response-template designs in ReSum and AgentFold. ReSum exposes summarization as a separate tool invocation that periodically converts a growing trajectory into a compact restartable state. AgentFold, by contrast, embeds context folding into the agent's own structured response format, so that the agent jointly reasons, updates its compressed state, and acts.

Table 4 further summarizes the operational flow of the two methods. Although both methods compress trajectories through natural-language abstraction, their control policies differ. ReSum is closer to periodic tool-mediated summarization, while AgentFold is closer to agent-controlled online state maintenance.

C.2 Algorithms for Pruning and Reduction

Pruning and reduction methods differ from abstractive summarization because they preserve selected source content exactly. This property is especially important for coding agents, where identifiers, file paths, syntax, and local dependencies often need exact retention. SWE-Pruner (Wang et al., 2026c) is a representative pruning-based method: it operates as a middleware between the coding agent and the environment, intercepts long code observations, and returns a shorter pruned observation to the agent.

The core scoring and aggregation procedure can be summarized as Algorithm 1. The notation is simplified to emphasize the compression operator rather than implementation-specific training details.

Algorithm 1 Task-aware line-level pruning in SWE-Pruner

Require: Code context C, pruning query or goal hint q, scoring model F_θ, threshold τ
Ensure: Pruned code context $\tilde{C}$
Split C into lines $L = \{l_1, \ldots, l_m\}$
Tokenize C into tokens $X = \{x_1, \ldots, x_n\}$
for each token $x_i \in X$ **do**
  Compute token relevance score $s_i = F_\theta(x_i, C, q)$
end for
for each line $l_j \in L$ **do**
  Let T_j be the set of tokens belonging to line l_j
  Aggregate token scores into a line score:

$$\bar{s}_j = \frac{1}{|T_j|} \sum_{x_i \in T_j} s_i$$

end for
Decode or threshold line scores to obtain keep/drop labels $y_j \in \{0, 1\}$
Construct $\tilde{C}$ by preserving lines with $y_j = 1$
return $\tilde{C}$

D Case Studies of Context Compression Failure Modes

This appendix provides three illustrative cases corresponding to the temporal failure taxonomy in Section 5. Each case follows the same pipeline: a task produces an original agent trajectory; the trajectory is compressed into a compact state; and the downstream agent fails after compression. The cases are constructed to clarify where the earliest causal error arises in the compression pipeline: before compression, during compression, or after compression.

D.1 F1 Example – Authorization State Dropped

Figure 5a illustrates an F1 failure in a file-management task. The agent is asked to delete all `.pdf` files under `~/downloads`. In the original trajectory, the agent logs in, obtains an `access_token`, attempts to inspect the target directory, and observes a `401 Unauthorized` response. The failed API call is not merely an incidental error: it reveals an operational rule that every subsequent file-system API call must explicitly pass the access token.

The crucial state before compression therefore consists of three linked pieces of information: the concrete token value, the API rule requiring the token, and the cause of the previous `401` error. How-

Method	Prompt / Template	Input	Output	Compression-specific instruction
ReSum (Wu et al., 2026c)	Context summarization prompt	The original question and the accumulated conversation history, which may include the agent's previous reasoning, tool calls, observations, and intermediate findings.	A compact summary block that condenses the previous trajectory into a restartable reasoning state.	The summary tool is instructed to extract task-relevant and reliable information from the history, preserve useful evidence and search progress, and avoid unsupported guesses or uncertain inferences.
ReSum (Wu et al., 2026c)	Summary-conditioned reasoning prompt	The original question together with the generated summary from previous exploration.	Continued reasoning and tool use conditioned on the generated summary.	The agent is instructed to treat the summary as condensed prior context, judge whether it is sufficient, and continue collecting evidence if the summary does not yet support a final answer.
AgentFold (Ye et al., 2025)	Structured agent response template	The task instruction, tool interface, current multi-scale state summaries, and latest high-fidelity interaction.	A structured response containing thinking, a folding directive, a brief explanation, and the next tool call/action.	Context management is made part of the agent's action space. Each step can update the compressed state while still preserving the latest interaction at high fidelity for immediate reasoning.
AgentFold (Ye et al., 2025)	Folding directive template	Existing state summaries plus the latest interaction.	A JSON-like folding command specifying which span should be folded and what summary should replace it.	The directive supports two modes: granular condensation, which folds a single latest step while preserving useful information, and deep consolidation, which folds several steps into a coarser summary when they complete a subtask and their intermediate details are no longer critical for further task solving.

Table 3: Prompt and response-template details for summarization and abstraction methods. ReSum uses explicit summarization and summary-conditioned reasoning prompts, whereas AgentFold embeds context folding into the agent's structured response format.

ever, the compression controller makes a wrong pre-compression retention decision. It treats the token, the failed API-call details, and the token-passing rule as disposable low-level execution noise, while preserving only the coarse task objective and a vague note that authorization failed.

The resulting compressed state is insufficient for correct continuation. After compression, the downstream agent still knows that it should delete PDFs under the target directory, but it no longer knows how to perform authorized API calls. It therefore repeats unauthorized calls and wastes steps on unnecessary exploration.

This is an F1 failure because the earliest causal error occurs before the compression operation itself. The problem is not that the summarizer distorted selected content, nor that a saved memory could not be retrieved later. Rather, the controller incorrectly decided that essential operational state was not worth preserving. The repeated `401` errors are downstream symptoms of this pre-compression decision error.

D.2 F2 Example – Uncertainty Loss

Figure 5b illustrates an F2 failure in a research-agent task. The agent is asked to assess whether Material A shows stable room-temperature superconductivity. The original trajectory contains heterogeneous evidence: one source reports a resistance drop near 295K, another reports no Meissner evidence, a third source reports failed reproduction, and a database entry marks the claim as unverified with conflicting reports. The agent's intermediate conclusion is therefore cautious: the evidence is inconclusive.

Unlike the F1 case, the relevant evidence has been collected and is available to the compression module. The failure occurs during compression. When the trajectory is condensed, the compression module collapses the evidence balance into a sim-

Method	Trigger	Compression process	Resulting state	Main risk
ReSum (Wu et al., 2026c)	Periodic or budget-driven summarization during long-horizon search.	The agent accumulates a ReAct-style trajectory until summarization is invoked. The summarizer rewrites the prior history into a compact state, after which the agent continues from the original question plus the generated summary rather than the full history.	A compact, restartable reasoning state that retains prior discoveries, useful clues, and unresolved directions.	The summary may omit evidence, over-compress uncertainty, or fail to preserve fine-grained dependencies needed in later search.
AgentFold (Ye et al., 2025)	Agent-initiated folding at each step.	At each step, the agent keeps the most recent interaction as high-fidelity working memory and issues a folding directive that either condenses a single step into a fine-grained summary or consolidates multiple steps into a coarser summary.	A multi-scale memory state consisting of fine-grained local summaries and coarser strategic summaries.	The agent may fold too aggressively, consolidate heterogeneous intermediate steps, or abstract away details that later become necessary.

Table 4: Operational flow of ReSum and AgentFold. Both compress trajectories into executable summaries, but they differ in whether summarization is invoked as an external tool or internalized as an agent action.

plified statement that Material A shows a 295K transition, suggesting possible room-temperature superconductivity. At the same time, the negative and uncertainty-bearing evidence is weakened or dropped: the lack of Meissner evidence, failed reproduction, unverified database status, and overall inconclusiveness no longer strongly constrain the compressed state.

This is an F2 failure because the compression process changes the epistemic status of the trajectory. The original evidence supports an inconclusive assessment, but the compressed state makes the claim appear more suggestive and better supported than it actually is. The downstream agent then overstates the evidence and produces an overconfident, factually misleading final answer (Shi et al., 2026d; Wang et al., 2025c).

The key distinction from F1 is that the information was not excluded by a prior retention decision. Instead, the evidence entered the compression step but was transformed into a less faithful representation. Thus, the earliest causal error lies inside the compression operation itself.

D.3 F3 Example – Wrong Retrieval

Figure 5c illustrates an F3 failure in a web-research task. The agent must find the CEO of Microsoft in 2024. In the original trajectory, it searches for "Microsoft CEO 2024", opens an official Microsoft source, reads the leadership page, and observes that Satya Nadella was appointed Chief Executive Officer of Microsoft on February 4, 2014. This correct evidence is then summarized and externalized as memory item `M123`, which stores the official Microsoft leadership evidence for the 2024 CEO.

Unlike the F2 case, the relevant information is not lost or distorted during compression. The compression operation successfully externalizes the correct evidence. However, the compressed active context does not contain the full content of `M123`; it contains the task, an external memory pointer, and an access mechanism indicating that the evidence should be retrieved when needed. The correct memory therefore exists, but it must re-enter the active context through a post-compression access step.

The failure occurs after compression. Instead of directly expanding the correct pointer `M123`, the agent issues the query "Microsoft CEO 2024". The retrieval mechanism returns a different memory item, `M087`, which is topically related but temporally mismatched: it states that Steve Ballmer was Microsoft CEO in 2013 based on an older article. The downstream agent integrates this wrong retrieved chunk into the active context and answers Steve Ballmer.

This is an F3 failure because the correct information has been retained, but it is not correctly accessed at the decision point. The earliest causal error is neither a pre-compression decision error nor an in-compression distortion. Rather, it is a post-compression access failure: the correct memory item `M123` exists, but the retrieval or expansion mechanism surfaces `M087` instead. The final answer is wrong because the agent reasons from the wrong retrieved memory, not because the correct evidence was never stored.

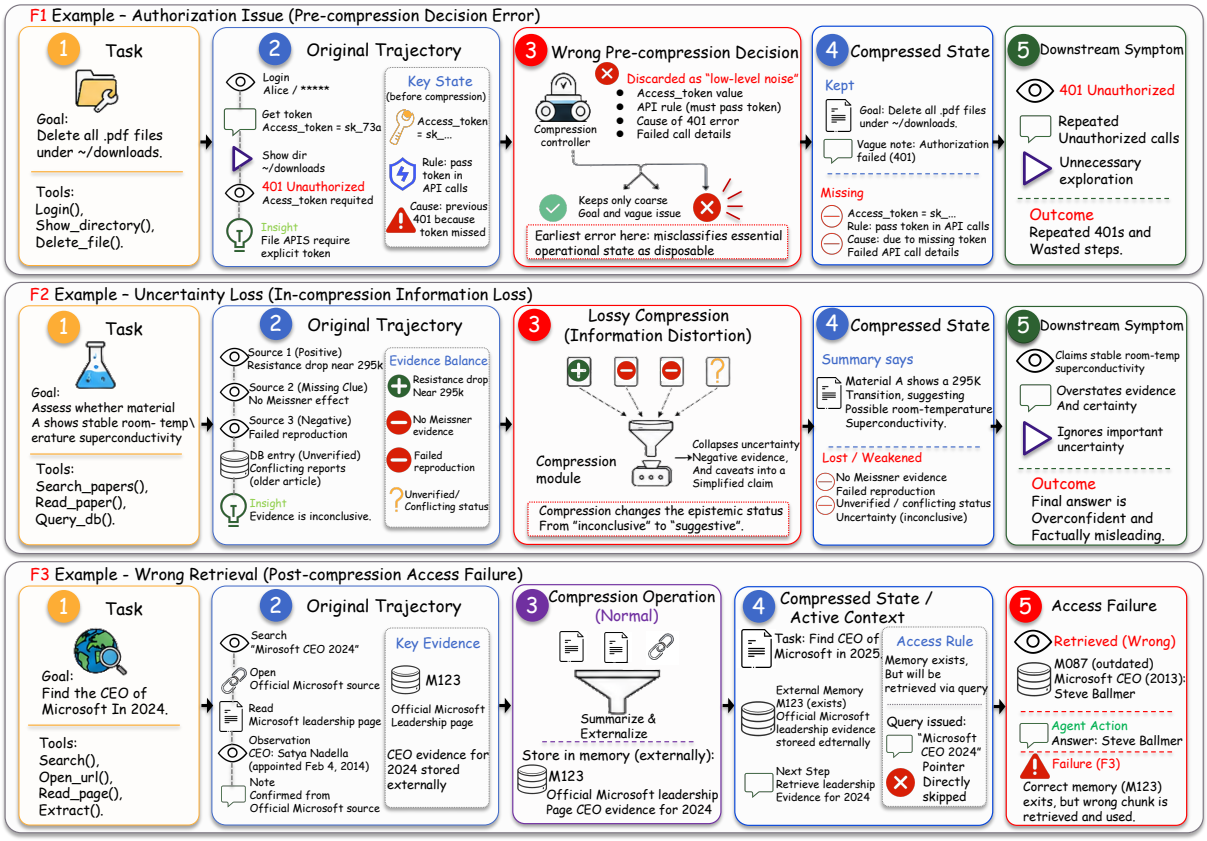


Figure 5: Three illustrative cases of context compression failure modes. F1 shows a pre-compression decision error, where the compression controller misclassifies authentication state and API-call details as low-level noise before summarization. F2 shows in-compression information loss, where a compression module distorts the evidential balance by weakening uncertainty, negative evidence, and caveats. F3 shows a post-compression access failure, where the correct memory item exists after compression but is not correctly expanded or retrieved, causing the agent to use an outdated wrong chunk.

E Discussion & Future Directions

The Compression–Retrieval–Memory Boundary In current agent systems, compression, retrieval, and external memory are often combined, but they serve different functions and have different recoverability properties. A more principled framework is needed to determine when information should be compressed, when it should be stored externally, and when it should remain in active context (Shi et al., 2026a).

Learning to Compress Recoverably Most existing learned policies optimize task success and token efficiency, but these objectives do not directly capture whether compressed information can be reconstructed when needed. Future work should incorporate recoverability and dependency preservation into both training objectives and evaluation.

Compression in Multi-Agent Systems When context is shared across agents, compression becomes a communication and coordination problem rather than a local memory optimization problem. This setting raises new questions about what should be summarized, what should be shared verbatim, and how compression errors propagate across agents (Zhang et al., 2026a,b).

End-to-End Training under Context Budgets Compression is usually treated as a separate module from planning and tool use, even though these decisions are tightly coupled in long-horizon agents. Joint optimization under explicit budget constraints may better align compression behavior with downstream task performance (Xiong et al., 2025; Wang et al., 2024b).

Domain-Specific Compression Methods As shown in § 6, coding, web/GUI, and research agents impose different requirements on structural fidelity, recoverability, and evidence precision. This suggests that domain-tailored compression strategies will be important in addition to general-

Dimension	Proposed Measurement / Formula	Example Baseline
D	$D_{\text{density}} = \frac{\text{count}(A_{\text{preserved}})}{\text{tokens}(C_{\text{compressed}})}$	Context Codec (Trukhina and Vashkelis, 2026)
R	$R@k = \frac{\text{count}(\text{Relevant} \cap \text{Retrieved}_k)}{\text{count}(\text{Relevant})}$	MEM1 (Zhou et al., 2025b)
P	$Hall_{QA} = \frac{N_{\text{hallucinated}}}{N_{\text{total}}}$; $\text{PropagationRate} = \frac{\text{Hall}_{QA}}{\text{Hall}_E}$	HaluMem (Chen et al., 2026a)
O	$O_{\text{ctx}} = \frac{\sum_{t=1}^{T} \text{tokens}(C_t^{\text{method}})}{\sum_{t=1}^{T} \text{tokens}(C_t^{\text{base}})}$	AgentFold (Ye et al., 2025)

Table 5: Dimensions and Proposed Measurements.

purpose methods (Lin et al., 2026a).

Standardized Benchmarks and Reproducibility Existing work differs widely in agent backbone, task set, budget, and compressed target, which makes direct comparison difficult (Zhang et al., 2026d; Chen et al., 2026d; Fang et al., 2025b). Future benchmarks should report not only task success and token savings, but also information density, recoverability, error propagation, and controller overhead under controlled settings.

We have systematically surveyed how existing works already operationalize these four dimensions, and summarize them in Table 5. To further demonstrate how this multi-dimensional perspective reveals the different evaluation focuses of current methods, we add the following mapping in Table 6.

Method	D	R	P	O
AgentFold	✓	—	—	✓
COMPASS	—	✓	✓	—
LongMemEval	—	✓	—	—
Context Codec	✓	✓	—	—

Table 6: Method Comparison

Clarification on the Weighted Score We acknowledge that combining these dimensions into a single weighted score may not be universally applicable. It is difficult to propose a unified framework for these coefficients, since their relative importance varies drastically across different tasks and constraints. Therefore, we position D/R/P/O as a flexible evaluation checklist rather than a rigid quantitative metric., leaving the exploration of specific coefficient weightings to future benchmark designs.

Formalization Distinction from ACON (Kang et al., 2025) We acknowledge that both works share a conceptual foundation in treating compression as a form of state approximation. While there are conceptual similarities, we would like to respectfully clarify the key differences between our formulation and ACON's, which stem from our distinct research goals:

- **Formal Object:** ACON formalizes context compression with the goal of *optimizing a specific compressor* (or guidelines) under a reward-cost trade-off. In contrast, our Section 3 aims to define an *ideal compressed state* that preserves necessary information for future agent behavior. Our formulation serves as a general abstraction designed to analyze and compare various existing compression operators, rather than to train a new one.
- **Sufficiency Criterion:** In ACON, state preservation is operationalized empirically through downstream task reward and context cost. In our formulation, sufficiency is defined theoretically by the behavioral gap ($\Delta_\tau \leq \epsilon$) over future decisions. This behavioral framing is specifically designed to allow us to ground and categorize the various failure risks (F1-F3) observed across different methods.

ID	Method	Author & Year	When	What	How	Domain
1	WebAgent	Gur et al., 2024	System-Controlled	Observation	Summarization & Abstraction	
2	PAL-UI	Liu et al., 2025f	Agent-Controlled	Trajectory	Externalization & Retrieval	
3	LCoW	Lee et al., 2025	Learned	Observation	Summarization & Abstraction	
4	Sculptor	Li et al., 2025a	Agent-Controlled	Memory State	Truncation & Masking / Abstraction	
5	SWE-Pruner	Wang et al., 2026c	Learned	Observation	Pruning & Reduction	
6	AgentOCR	Feng et al., 2026	Agent-Controlled	Trajectory	Representation-Level	
7	Step-DeepResearch	Hu et al., 2025a	System-Controlled	Observation	Externalization & Retrieval	
8	MCP-Bench	Wang et al., 2025d	System-Controlled	Observation	Summarization & Abstraction	
9	PAACE	Yuksel, 2025	Learned	Plan & Reasoning	Pruning & Abstraction	
10	WebWeaver	Li et al., 2025b	Agent-Controlled	Memory State	Externalization & Retrieval	
11	The Complexity Trap	Lindenbauer et al., 2025	System-Controlled	Observation	Truncation & Masking	
12	WebClipper	Wang et al., 2026a	Learned	Trajectory	Pruning & Reduction	
13	AOI	Bai et al., 2026	System-Controlled	Memory State	Summarization & Abstraction	
14	COMPASS	Wan et al., 2025	External Controller	Plan & Reasoning	Summarization & Abstraction	
15	ACC	Bousetouane, 2026	External Controller	Memory State	Summarization & Abstraction	
16	BACM	Xiao et al., 2025	Agent-Controlled	Trajectory	Summarization & Abstraction	
17	Q-KVComm	Kriuk and Ng, 2025	System-Controlled	Representation-Level	Representation-Level	
18	Visual MAD	Wu et al., 2026a	System-Controlled	Trajectory	Representation-Level	
19	SCMA	Chen et al., 2026c	Learned	Plan & Reasoning	Pruning & Reduction	
20	CSIM	Liu et al., 2025d	Learned	Plan & Reasoning	Summarization & Abstraction	
21	ACON	Kang et al., 2025	Learned	Trajectory & Observation	Summarization & Abstraction	
22	SAC	Liu et al., 2026b	System-Controlled	Representation-Level	Representation-Level	
23	PoC	Zhao et al., 2026b	External Controller	Trajectory & Observation	Pruning & Abstraction	
24	SWE-AGILE	Lian et al., 2026	Learned	Plan & Reasoning	Truncation & Abstraction	
25	LoCoEval Framework	Liu et al., 2026c	System-Controlled	Memory State	Externalization & Retrieval	
26	LongSeeker	Lu et al., 2026	Agent-Controlled	Trajectory	Pruning & Abstraction & Retrieval	
27	Context-Folding	Sun et al., 2025	Agent-Controlled	Trajectory	Summarization & Abstraction	

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ID	Method	Author	When	What	How	Domain
28	ContextWeaver	Wu et al., 2026d	External Controller	Memory State	Pruning & Reduction	
29	HIAGENT	Hu et al., 2025b	Agent-Controlled	Plan & Reasoning	Summarization & Abstraction	
30	ProMem	Yang et al., 2026a	Agent-Controlled	Memory State	Summarization & Abstraction	
31	OCR-Memory	Li et al., 2026a	External Controller	Memory State	Externalization & Retrieval	
32	AgentDiet	Xiao et al., 2025	System-Controlled	Trajectory	Pruning & Reduction	
33	AgentProg	Tian et al., 2026	System-Controlled	Memory State	Pruning & Reduction	
34	GCC	Wu et al., 2026b	Agent-Controlled	Memory State	Externalization & Retrieval	
35	TACO	Ren et al., 2026	Learned	Observation	Pruning & Reduction	
36	CodeOCR	Shi et al., 2026b	System-Controlled	Observation	Representation-Level	
37	AttentionRAG	Fang et al., 2025a	System-Controlled	Observation	Pruning & Reduction	
38	LongCodeZip	Shi et al., 2025	System-Controlled	Observation	Pruning & Reduction	
39	ReSum	Wu et al., 2026c	System-Controlled	Trajectory	Summarization & Abstraction	
40	Focus	Verma, 2026	Agent-Controlled	Trajectory	Pruning & Reduction	
41	AgentFold	Ye et al., 2025	Agent-Controlled	Trajectory	Summarization & Abstraction	
42	CAT	Liu et al., 2025b	Agent-Controlled	Trajectory	Summarization & Abstraction	
43	MEM1	Zhou et al., 2025b	Learned	Memory State	Summarization & Abstraction	
44	ContextEvolve	Su et al., 2026	External Controller	Memory State	Summarization & Abstraction	
45	TRACE	Min et al., 2026	System-Controlled	Trajectory	Summarization & Abstraction	
46	ACM	Li et al., 2026b	Agent-Controlled	Trajectory & Observation	Externalization & Retrieval / Summarization & Abstraction	
47	Agentic Context Management	Dadhich, 2026	External Controller	Memory State & Active Context	Externalization & Retrieval & Summarization & Abstraction	
48	PACMS	Ghulyani et al., 2026	System-Controlled	Pooled Conversation Turns & Memory Entries & Tool Outputs	Pruning & Reduction	
49	Telegraph English	Bei et al., 2026	System-Controlled	Observation & Document Passages	Summarization & Abstraction	
50	C-DIC	Jung et al., 2026	Agent-Controlled	Trajectory	Representation-level & Externalization & Retrieval	
51	AdaCoM	Yi et al., 2026	Learned	Trajectory	Pruning & Reduction & Summarization & Abstraction	

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ID	Method	Author	When	What	How	Domain
52	ZipRL	Hu et al., 2026b	Learned	Trajectory & Observation	Summarization & Abstraction	
53	Context Codec	Trukhina and Vashkelis, 2026	External Controller	Plan & Reasoning	Summarization & Abstraction	
54	SeCo	Tang et al., 2026a	Learned	Representation-Level	Representation-Level	
55	ConAct	Liu et al., 2026a	Agent-Controlled	Trajectory	Summarization & Abstraction	
56	CoMem	Zhang et al., 2026f	External Controller	Trajectory & Memory State	Externalization & Retrieval	
57	RoMeRL	Yang et al., 2026b	Learned	Memory State	Pruning & Reduction	
58	Router-Mem	Lin et al., 2026b	Learned External Controller	Memory State & Trajectory	Externalization & Retrieval	
59	MESA	Zhao et al., 2026a	Learned	Memory State	Externalization & Retrieval & Pruning & Reduction	
60	CommitKV	Huang et al., 2026	System-Controlled	Representation-Level & Trajectory	Pruning & Reduction	
61	SeDeM	Haghifam et al., 2026	Learned	Representation-Level	Representation-Level & Retrieval	
62	Adaptive Context Compression	Yang et al., 2026c	Learned	Memory State & Trajectory	Pruning & Reduction & Summarization & Abstraction	
63	GRC	Miao et al., 2026	System-Controlled	Representation-Level & Memory State	Representation-Level & Externalization & Retrieval	
64	AutoWorld-Builder	Chen et al., 2026b	System-Controlled	Trajectory & Memory State	Summarization & Abstraction	

Table 7: Full Taxonomy Coordinates. Domain icons: Coding Agents, Web Agents, GUI Interaction Agents, Deep-Search Agents, Research Agents, General Agents.

Method	Open-source link	Year
SWE-Pruner (Wang et al., 2026c)	https://github.com/Ayanami1314/swe-pruner	2026
Step-DeepResearch (Hu et al., 2025a)	https://github.com/stepfun-ai/StepDeepResearch	2025
MCP-Bench (Wang et al., 2025d)	https://github.com/Accenture/mcp-bench	2025
WebWeaver (Li et al., 2025b)	https://github.com/Alibaba-NLP/DeepResearch	2025
The Complexity Trap (Lindenbauer et al., 2025)	https://github.com/JetBrains-Research/the-complexity-trap	2025
WebClipper (Wang et al., 2026a)	https://github.com/AQ-MedAI/AntAFu-DeepResearch	2026
ACON (Kang et al., 2025)	https://github.com/microsoft/acon	2025
SAC (Liu et al., 2026b)	https://github.com/lx-Meteors/SAC	2026
SWE-AGILE (Lian et al., 2026)	https://github.com/KDEGroup/SWE-AGILE	2026
HIAGENT (Hu et al., 2025b)	https://github.com/HiAgent2024/HiAgent	2025
AgentProg (Tian et al., 2026)	https://github.com/MobileLLM/AgentProg	2026
TACO (Ren et al., 2026)	https://github.com/multimodal-art-projection/TACO	2026
CodeOCR (Shi et al., 2026b)	https://github.com/YerbaPage/CodeOCR	2026
LongCodeZip (Shi et al., 2025)	https://github.com/YerbaPage/LongCodeZip	2025
AgentFold (Ye et al., 2025)	https://github.com/Alibaba-NLP/DeepResearch	2025
MEM1 (Zhou et al., 2025b)	https://github.com/MIT-MI/MEM1	2025
TRACE (Min et al., 2026)	https://github.com/nokia-applied-research/Trace	2026
ACM (Li et al., 2026b)	https://github.com/lixiaochuan2020/agentic-context-management	2026
ZipRL (Hu et al., 2026b)	https://github.com/huzhexin/ZipRL	2026
ConAct (Liu et al., 2026a)	https://github.com/kwai/MemGUI-Agent	2026
CoMem (Zhang et al., 2026f)	https://github.com/horizon-llm/CoMem	2026
RoMeRL (Yang et al., 2026b)	https://github.com/YOUNG-fnxm/RoMeRL	2026
ACC (Yang et al., 2026c)	https://github.com/usail-hkust/EvoDS	2026

Table 8: Open-source methods included in this survey.

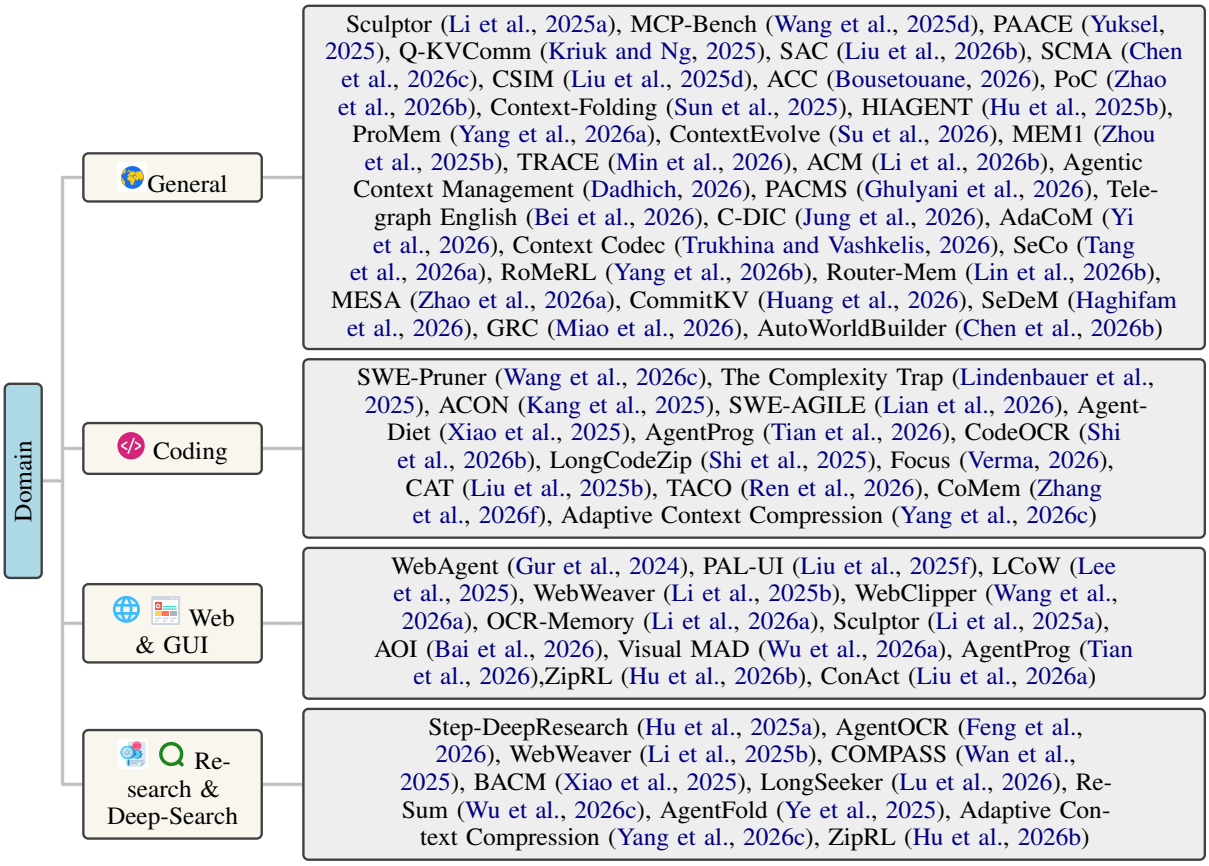


Figure 6: A taxonomy of context management methods from the perspective of domain.

Benchmark	Source & year	Open Source
SWE-bench Verified (Jimenez et al., 2024)	ICLR 2024	https://huggingface.co/datasets/SWE-bench/SWE-bench_Verified
GAIA (Mialon et al., 2023)	arXiv 2023	https://huggingface.co/datasets/gaia-benchmark/GAIA
BrowseComp (Wei et al., 2025)	arXiv 2025	https://github.com/openai/simple-evals
BrowseComp-Plus (Chen et al., 2025d)	ACL 2026	https://huggingface.co/datasets/Tevatron/browsecomp-plus
BrowseComp-ZH (Zhou et al., 2025a)	arXiv 2025	https://huggingface.co/datasets/PALIN2018/BrowseComp-ZH
Mind2Web (Deng et al., 2023)	NeurIPS 2023	https://huggingface.co/datasets/osunlp/Mind2Web
Multimodal-Mind2Web (Zheng et al., 2024)	ICML 2024	https://huggingface.co/datasets/osunlp/Multimodal-Mind2Web
HLE (Phan et al., 2025)	arXiv 2025	https://huggingface.co/datasets/cais/hle
AppWorld (Trivedi et al., 2024)	ACL 2024	https://github.com/StonyBrookNLP/appworld
OfficeBench (Wang et al., 2024c)	arXiv 2024	https://github.com/zlwang-cs/OfficeBench
HotpotQA (Yang et al., 2018)	EMNLP 2018	https://hotpotqa.github.io/
Natural Questions (Kwiatkowski et al., 2019)	TACL 2019	https://github.com/google-research-datasets/natural-questions
TriviaQA (Joshi et al., 2017)	ACL 2017	http://nlp.cs.washington.edu/triviaqa/
GSM8K (Cobbe et al., 2021)	arXiv 2021	https://github.com/openai/grade-school-math
MATH / MATH-500 (Hendrycks et al., 2021)	NeurIPS 2021	https://github.com/hendrycks/math
SWE-bench Lite (Jimenez et al., 2024)	ICLR 2024	https://huggingface.co/datasets/princeton-nlp/SWE-bench_Lite
ALFWorld (Shridhar et al., 2021)	ICLR 2021	https://github.com/alfworld/alfworld
WebShop (Yao et al., 2022)	NeurIPS 2022	https://github.com/princeton-nlp/webshop
AndroidWorld / AW-Extend (Rawles et al., 2024)	ICLR 2025	https://github.com/google-research/android_world
DeepResearch Bench (Du et al., 2025a)	arXiv 2025	https://github.com/Ayanami0730/deep_research_bench
TerminalBench (TB 1.0 / 2.0) (Merrill et al., 2026)	arXiv 2026	https://github.com/harbor-framework/terminal-bench
MRQA (Fisch et al., 2019)	EMNLP WS 2019	https://huggingface.co/datasets/mrqa
CodeXGLUE (Lu et al., 2021)	NeurIPS / FSE 2021	https://github.com/microsoft/CodeXGLUE
AIOpsLab (Chen et al., 2025c)	MLSys 2025	https://microsoft.github.io/AIOpsLab/
Loghub (Zhu et al., 2023)	ISSRE 2023	https://github.com/logpai/loghub
LongBench (Bai et al., 2024)	MLSys 2024	https://huggingface.co/datasets/THUDM/LongBench
LongBench V2 (Bai et al., 2025)	ACL 2025	https://longbench2.github.io
BABILong (Kuratov et al., 2024)	NeurIPS 2024	https://github.com/booydar/babilong
ADRS Benchmark (Cheng et al., 2025)	arXiv 2025	https://github.com/UCB-ADRS/ADRS
WorkArena (Drouin et al., 2024)	ICML 2024	https://github.com/ServiceNow/WorkArena
MiniWoB (Shi et al., 2017)	ICML 2017	https://miniwob.farama.org/
MiniWoB++ (Liu et al., 2018)	ICLR 2018	https://miniwob.farama.org/
NeedleBench (Li et al., 2024c)	arXiv 2024	https://github.com/open-compass/opencompass
NIAH (Liu et al., 2024)	arXiv 2023	https://github.com/open-compass/opencompass
HaluMem (Chen et al., 2025a)	arXiv 2025	https://huggingface.co/datasets/IAAR-Shanghai/HaluMem
LongMemEval (Wu et al., 2025)	ICLR 2025	https://huggingface.co/datasets/xiaowu0162/longmemeval-cleaned
DevEval (Li et al., 2024b)	ACL Findings 2024	https://github.com/seketeam/DevEval
WideSearch (Wong et al., 2025)	ICLR 2026	https://huggingface.co/datasets/ByteDance-Seed/WideSearch
CompileBench / CRUST-Bench (Khatry et al., 2025)	COLM 2025	https://github.com/QuesmaOrg/CompileBench
GovReport (Huang et al., 2021)	NAACL 2021	https://gov-report-data.github.io/
SummScreenFD (Chen et al., 2022)	ACL 2022	https://gov-report-data.github.io/

Table 9: Representative datasets and benchmarks for agent context management.